



# UNIVERSITY OF EDUCATION WINNEBA

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## DEPARTMENT OF ICT EDUCATION

### B.Sc. ICT EDUCATION

**END OF SECOND SEMESTER EXAMINATION,  
2020/2021**

**MATD 122 – CALCULUS**

**DO NOT TURN OVER UNTIL YOU ARE TOLD TO  
BEGIN**

Time allowed: **TWO** hours

**DATE: SEPTEMBER, 2021**

**SECTION A**  
**40 MARKS**  
**INSTRUCTION: ATTEMPT ALL QUESTIONS**

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1. Find the domain of the function  $F(x) = \frac{x+2}{x^2+3x-4}$
- A.  $\{x|x \in R, x \neq 4 \text{ and } -1\}$       B.  $\{x|x \in R, x \neq -4 \text{ and } 1\}$   
C.  $\{x|x \in R, x = -4 \text{ and } 1\}$       D.  $\{x|x \in R, x = 4 \text{ and } -1\}$
2. Which of the following limits represent the derivative of a function from definition?
- A.  $\lim_{h \rightarrow 0} \frac{f(x)-f(x+h)}{h}$       B.  $\lim_{h \rightarrow \infty} \frac{f(x+h)-f(x)}{h}$   
C.  $\lim_{h \rightarrow \infty} \frac{f(x+h)+f(x)}{h}$       D.  $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$
3. What is the answer when you divide  $4x^3 + 13x + 33$  by  $2x + 3$ ?
- A.  $2x^2 + 3x + 11$       B.  $2x^2 - 3x + 11$   
C.  $2x^2 + x + 11$       D.  $2x^2 - x + 11$
4. What is the result of simplifying  $\frac{1}{1-\cos\theta} + \frac{1}{1+\cos\theta}$ ?
- A.  $2\operatorname{cosec}^2\theta$       B.  $2\sec^2\theta$       C.  $2\cos^2\theta$       D.  $2\sin^2\theta$
5. If  $f(x) = \sqrt{x}$  and  $g(x) = \sqrt{1-x}$ , find the domain of  $f(x)/g(x)$
- A.  $\{x|x \in R, x = 0 \text{ and } 1\}$       B.  $\{x|x \in R, 0 < x \leq 1\}$   
C.  $\{x|x \in R, x \neq 0\}$       D.  $\{x|x \in R, 0 \leq x < 1\}$
6. Find the inverse  $h^{-1}(x)$  of the function  $h(x) = \frac{x+3}{2x-1}$ , where  $x \neq 1/2$
- A.  $\frac{x+3}{2x-1}$       B.  $\frac{x-3}{2x+1}$       C.  $\frac{2x-1}{x+3}$       D.  $\frac{2x+1}{x-3}$
7. If  $f(x) = 2x^2 + 3$  and  $h(x) = \frac{x+1}{x-1}$ , find  $h \circ f(x)$
- A.  $\frac{x^2+3}{x^2+2}$       B.  $\frac{x^2-3}{x^2-2}$       C.  $\frac{x^2-2}{x^2-1}$       D.  $\frac{x^2+2}{x^2+1}$
8. If  $f(x) = \frac{u(x)}{v(x)}$ , what is  $f'(x)$

A.  $\frac{u(x) \cdot v'(x) - v(x) \cdot u'(x)}{[v(x)]^2}$

B.  $\frac{v(x) \cdot u'(x) - u(x) \cdot v'(x)}{[v(x)]^2}$

C.  $\frac{u(x) \cdot v'(x) + v(x) \cdot u'(x)}{[v(x)]^2}$

D.  $\frac{v(x) \cdot u'(x) + u(x) \cdot v'(x)}{[v(x)]^2}$

9. Evaluate  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

- A. the limit does not exist    B. 2    C. 0    D. -1

10. Evaluate  $\lim_{x \rightarrow 1} f(x)$  if  $f(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ 3 & \text{if } x = 1 \end{cases}$

- A. 0    B. 1    C. the limit does not exist    D. 2

11. find  $\frac{dy}{dx}$  if  $y = \frac{x+1}{x-2}$

A.  $\frac{-3}{(x-2)^2}$

B.  $\frac{3}{(x-2)^2}$

C.  $\frac{-2}{(x-3)^2}$

D.  $\frac{2}{(x-3)^2}$

12. Which of the following is an odd function?

A.  $f(x) = \cos x$

B.  $f(x) = |x|$

C.  $f(x) = \sin x$

D.  $f(x) = 2x^4$

13. Evaluate  $\lim_{x \rightarrow -\infty} \frac{x(2x-1)^2}{(3x+1)(x-2)^2}$

A.  $\frac{4}{3}$

B.  $\frac{3}{4}$

- C. the limit does not exist    D. 0

14. Evaluate  $\int_1^2 \sqrt{x} \, dx$

A.  $4\frac{1}{3}$

B. 2

C.  $4\frac{2}{3}$

D. 1

15. Evaluate  $\int (x^4 - 2x^2 + 1) \, dx$

A.  $\frac{x^5}{5} + \frac{2x^3}{3} + x + c$

B.  $\frac{x^5}{5} - \frac{2x^3}{3} + x + c$

C.  $\frac{x^4}{4} - \frac{x^2}{2} + x + c$

D.  $\frac{x^4}{4} - \frac{2x^2}{2} + x + c$

16. Find the gradient of the curve  $y = 2x^3 + 3x^2 - 2x - 3$  at the point P (1, 0)

- A. -10      B. 1      C. 10      D. 0

17. Evaluate  $\lim_{x \rightarrow 1} f(x)$ , where  $f(x) = \begin{cases} 2 - x & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ x & \text{if } x > 1 \end{cases}$

- A. 0    B. 1    C. the limit does not exist    D. 2

18. Simplify  $(1 - \cos^2 \theta) \sec^2 \theta$

- A.  $\tan^2 \theta$       B.  $\cot^2 \theta$       C.  $\sin^2 \theta$       D.  $\operatorname{cosec}^2 \theta$

19. if  $y = 3x^4 + 2x^3 - 5$  evaluate  $D^3 y$

- A.  $3x - 5$       B.  $3x + 2$       C.  $12x + 6$       D.  $54x + 12$

20. Evaluate  $\lim_{x \rightarrow \infty} \frac{2x+1}{x-1}$

- A. 2      B. 1      C. the limit does not exist    D. 0

21. If  $f(x) = x^2 \sin x$ , find  $f'(x)$

- A.  $2x \cos x + x^2 \sin x$     B.  $\cos x + \sin x$     C.  $2 \cos x + \sin x$     D.  $x^2 \cos x + 2x \sin x$

22. Find the remainder if  $x^2 - 5x + 1$  is divided by  $x - 2$

- A. -5      B. -4      D. -3      E. -2

23. Express  $\frac{x^2+x+2}{x-1}$  in the form  $A(x) + \frac{B}{x-1}$

- A.  $x + 3 + \frac{4}{x-1}$       B.  $x + 2 + \frac{4}{x-1}$   
C.  $x + 5 + \frac{4}{x-1}$       D.  $x + 1 + \frac{4}{x-1}$

24. Find the indefinite integral.

$$\int (6 - 6x^3 + 2x^7) dx$$

- a.  $\frac{1}{4}x^8 - \frac{3}{2}x^4 + 6x + C$   
b.  $6x - 6x^4 + 2x^8 + C$

c.  $\frac{1}{4}x^7 - \frac{3}{2}x^3 + 6x + C$

d.  $6x - 18x^2 + 14x^6 + C$

25. Find the derivative of the function using the definition of derivative.

$$g(t) = \frac{1}{\sqrt{6t}}$$

a.  $g'(t) = -\frac{1}{2\sqrt{6t^3}}$

b.  $g'(t) = \frac{1}{2\sqrt{6t^3}}$

c.  $g'(t) = \frac{1}{\sqrt{36t^3}}$

d.  $g'(t) = -\frac{1}{\sqrt{216t^3}}$

26. Use the alternative form of the derivative to find the derivative of the function  $f(x) = \frac{5}{x^3}$  at  $x = 3$ .

a.  $f'(3) = \frac{5}{27}$

b.  $f'(3) = -\frac{5}{27}$

c.  $f'(3) = \frac{5}{81}$

d.  $f'(3) = -\frac{5}{3}$

27. Use the definition of the derivative to find  $f''$ .

$$f(x) = x^3 - 9x^2$$

a.  $f(x) = 6x - 18$

b.  $f(x) = 3x - 18$

c.  $f(x) = 6$

d.  $f(x) = 18x$

28. Find the slope of the graph of the function at the given value.

$$f(x) = \frac{3}{x^2} \text{ when } x = 7$$

a.  $f'(7) = \frac{6}{343}$

b.  $f'(7) = \frac{3}{49}$

c.  $f'(7) = -\frac{6}{7}$

d.  $f'(7) = -\frac{6}{343}$

29. Given that  $\int_3^5 f(x) \, dx = -6$ , evaluate the following integral.

$$\int_3^5 -3f(x) \, dx$$

a.  $-18$

b.  $18$

c.  $-6$

d.  $6$

30. Find the derivative of the function

$$f(x) = -15x + 4$$

a.  $f'(x) = -15$

b.  $f'(x) = 4$

c.  $f'(x) = -15x$

d.  $f'(x) = -30x + 8$

31. Find the limit.

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 9}}{3x - 9}$$

a.  $-3$

b.  $3$

c. 9

d.  $\frac{1}{3}$

32. Give the domain and the range of the relation:

$$\{(11, -3), (2, -2), (2, 0), (6, 2), (18, 4)\}$$

- a. domain:  $\{-3, -2, 2, 4\}$ ; range:  $\{11, 6, 2, 18\}$
- b. domain:  $\{11, 6, 2, 18\}$ ; range:  $\{-3, -2, 2, 4\}$
- c. domain:  $\{11, 6, 2, 18\}$ ; range:  $\{-3, -2, 0, 2, 4\}$
- d. domain:  $\{-3, -2, 0, 2, 4\}$ ; range:  $\{11, 6, 2, 18\}$

33. Find the degree of the polynomial function  $f(x) = 5x - x^6 + \frac{5}{4}$

- a. 5
- b. 1
- c. -1
- d. 6

34. Find the domain of the rational function  $f(x) = \frac{x+8}{x^2-4}$

- a.  $\{x|x \neq -2, x \neq 2, x \neq -8\}$
- b.  $\{x|x \neq 0, x \neq 4\}$
- c.  $\{x|x \neq -2, x \neq 2\}$
- d. *all real numbers*

Determine whether the following are polynomials or not

35.  $g(x) = \frac{x^4-7}{x^6}$

- a. Yes
- b. No

36.  $h(x) = x^{3/2} - x^6 + 6$

- a. Yes
- b. No

37.  $y(x) = 4x^3 + 5x^2 - 4x^{-2} + 2$

- a. Yes
- b. No

Determine whether the given function is even, odd, or neither

38.  $f(x) = 5x^2 + x^4$

- a. Even
- b. Odd
- c. Neither

39.  $f(x) = -5x^5 + x^3$

- a. Odd
- b. Even
- c. Neither

40. Evaluate the piecewise function at the given value of the independent variable;

$$f(x) = \begin{cases} 3x + 1, & \text{if } x < -1 \\ -2x - 5, & \text{if } x \geq -1 \end{cases}; f(2)$$

- a.  $-9$
- b.  $-8$
- c.  $-3$
- d.  $1$



**SECTION B**  
**60 MARKS**

**INSTRUCTION: ANSWER TWO QUESTIONS FROM THIS SECTION. QUESTION ONE (1) AND ANY OTHER QUESTION.**  
**ALL QUESTION CARRY THE SAME MARKS**

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1. a. Differentiate from first principle

$$f(x) = \frac{1}{2-x} \quad 10 \text{ marks}$$

- b. Evaluate  $\lim_{x \rightarrow 0} \frac{\sqrt{2-x}-\sqrt{2}}{x}$  10 marks

- c. Simplify  $\frac{5-2\sqrt{3}}{\sqrt{3}-1}$  giving your answer in the form  $p+q\sqrt{3}$  where  $p$  and  $q$  are rational numbers 5 marks

- d. Find the value of the following limit 5 marks

$$\lim_{x \rightarrow 2} \left[ \frac{x^3 - 8}{x - 2} \right]$$

2. a. If  $f(x) = \begin{cases} 1-x^2 & \text{if } x < 1 \\ x-1 & \text{if } x > 1 \end{cases}$

5 marks

- i. Show that  $\lim_{x \rightarrow 1} f(x)$  exist and hence find the limit 5 marks

- ii. Sketch the graph of  $f(x)$

- b. If  $f(x) = \frac{x+2}{2x+1}$  and  $g(x) = \frac{x}{x-2}$  find and simplify the functions

5 marks

- i.  $f \circ g$

5 marks

- ii.  $g \circ f$

10 marks

- c. Find the domains of i above.

3. a. Show that  $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan \theta$

10 marks

- b. Express  $\frac{x^2+2x+6}{x-1}$  in the form  $A(x) + \frac{c}{x-1}$ , where  $A(x)$  is a function of  $x$  and  $c$  is a constants 5 marks

- c. show the if  $f(x) = \tan x$  the  $f'(x) = \sec^2 x$  10 marks

d. Find the value of the following limit

$$\lim_{x \rightarrow \infty} \left[ \frac{3x^2 + 7x - 1}{x^2 + 5} \right]$$

5 marks